What Shape Is a Reward?

How reward shape changes the feedback needed to recover it

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Abstract

Reward learning recovers a reward from feedback, but how much feedback it takes depends on what the reward looks like. We study five reward shapes in one 8×8 slippery gridworld: a sparse goal, a dense distance reward, a potential-shaped goal, two goals, and a goal with lava. We recover each from synthetic pairwise trajectory preferences (Bradley–Terry) or demonstrations (MaxEnt IRL), over 10 seeds and $N \in \{1, \ldots, 1000\}$. Everything is tabular, so regret and EPIC distance are computed exactly. Three hypotheses were fixed before the confirmatory run. **H1** (sparse and lava rewards need $\geq 3\times$ the preferences that dense needs) was not supported: sparse needed 3 pairs, the same as dense. **H2** (EPIC distance predicts regret better than raw reward correlation, most of all for shaped rewards) was supported: the Spearman correlations with regret were $\rho = 0.61$ against 0.54, and 0.46 against 0.05 on the shaped reward. **H3** (a myopic labeler leaves a regret floor that does not shrink with N) was not supported as stated, though the bias did leave large residual regret. Two unplanned observations stand out. Regret from preferences is not monotone in N for the dense and shaped rewards. And under the myopic labeler, regret is lowest at about ten pairs and then *rises* with more data. All labelers and parameters are simulated or invented and are labelled as such.

1 Introduction

Learning a reward from human feedback, whether comparisons [1, 2] or demonstrations [10], is now standard practice. It is well known that feedback identifies a reward only up to transformations such as potential shaping [6, 8], that reward distances should respect those invariances [4, 9], and that optimizing a learned proxy can backfire [3]. Less is known about a simpler question: *for a fixed environment, does the shape of the true reward change how much feedback it takes to recover it?*

We answer that question in a fully controlled setting. One gridworld, five ground-truth shapes, two feedback types, a well-specified reward model, exact metrics, ten seeds, and hypotheses stated before the confirmatory run. The setting is deliberately small. Every quantity is exact, so all variance comes from feedback sampling and can be measured.

Contributions. (i) Sample-complexity curves for reward recovery across five reward shapes and two feedback types (§3.1). (ii) A test of whether EPIC predicts downstream regret better than raw correlation, broken down by shape (§3.2). (iii) Misspecification ablations, covering labeler myopia and rationality, plus an overoptimization sweep (§3.3), and a shaping-for-speed check with Q-learning (§3.4). (iv) Open code that regenerates every number and figure with one command.

2 Setup

Environment (invented parameters). An 8×8 grid with four actions. With probability 0.9 the move goes as intended, otherwise in a uniformly random direction; walls keep the agent in place. There is no terminal state, and $\gamma = 0.95$. Trajectories have horizon $H = 20$ and uniform start states.

Reward shapes. Every ground truth is a table $R(s, a, s')$, rescaled to unit standard deviation over feasible transitions so that labeler noise means the same thing for every shape (Fig. 1): **sparse**, $+1$ for entering the goal $(7, 7)$; **dense**, $-d(s', g)/14$ with d the Manhattan distance; **shaped**, sparse $+\gamma\Phi(s') - \Phi(s)$ with $\Phi = -d(\cdot, g)$, which has the same optimal policy as sparse [6]; **multi**, $+1$ at $(7, 7)$ and $+0.6$ at $(0, 7)$; **lava**, $+1$ at the goal, -1 on six lava cells, and -0.02 per step.

Feedback. *Preferences:* each pair of trajectories shares a start state (as RLHF pairs share a prompt). The trajectories come from ϵ-greedy versions of the true optimal policy, with $\epsilon \sim U[0, 1]$ per trajectory. A Boltzmann labeler prefers τ_1 with probability $\sigma(\beta(G(\tau_1) - G(\tau_2)))$, where G is the γ-discounted return and $\beta = 1$. *Demonstrations:* trajectories from the finite-horizon soft-optimal policy at temperature 1.

Reward model and learners. $\hat{R}(s, a, s') = u(s) + w(s, a) + v(s')$, with 384 parameters. The class contains every ground truth, including the shaping term. Bradley–Terry is logistic regression on discounted feature counts (convex; L-BFGS, $\ell_2 = 10^{-3}$). MaxEnt IRL maximizes the demonstration likelihood through differentiable finite-horizon soft value iteration.

Metrics. *Normalized regret* $= (J(\pi^*) - J(\hat{\pi}))/(J(\pi^*) - J(\pi_{\text{unif}}))$, where $\hat{\pi}$ is greedy for $\hat{R}$, J is the exact true discounted return from a uniform start, and ties are broken uniformly (so a constant reward scores exactly 1). A reward is *recovered* when the mean regret is ≤ 0.05, and N^* is the smallest N on the grid where that holds. *EPIC* [4]: canonicalize both rewards with uniform $\mathcal{D}_S, \mathcal{D}_A$, then take the Pearson distance $\sqrt{(1 - \rho)/2}$ over the uniform product $S \times A \times S'$, computed exactly. *Raw Pearson*: the same distance without canonicalization, on the feasible-transition distribution.

Protocol. Tuning used seeds 100–104 only; every reported number uses seeds 0–9. In total there are 1400 reward-learning runs (about 125 CPU-minutes). Intervals are 95% percentile bootstraps over seeds. Spearman intervals come from a bootstrap that resamples whole seeds.

A pilot finding that shaped the design. In the pilot, the labeler summed rewards without discounting. Undiscounted returns identify a reward only up to γ=1 shaping, which is not policy-preserving at γ=0.95, and shaped-reward regret *grew* with N. We therefore matched the labeler's discount to the task's and shared start states. We report this because it is an easy trap.

Hypotheses (fixed before the confirmatory run). **H1**: sparse and lava need $\geq 3\times$ the preferences that dense needs to reach N^*. **H2**: across all runs, EPIC has a higher Spearman correlation with regret than raw Pearson, with the largest gap on the shaped reward. **H3**: a myopic labeler (discount 0.8; the model still assumes 0.95) leaves a regret floor that does not shrink with N, largest for sparse and lava.

3 Results

3.1 How much feedback until recovery?

Figure 2 shows regret and EPIC against N. From preferences, N^* was 3 (sparse), 3 (dense), 10 (shaped), 3 (multi) and 10 (lava). From demonstrations it was 1, 30, 100, 3 and 3. **H1 is not supported**: the sparse/dense ratio is $1.0\times$ and the lava/dense ratio $3.3\times$. In this setting, a sparse goal is easy to recover from comparisons. Our reading, untested: the compared trajectories are often near-optimal, preferred ones end at the goal, and that alone is enough for a greedy policy. The uninformed controls (a random reward) score regret between 0.88 and 1.04.

Two patterns we did not predict. *First*, preference-based regret is not monotone in N for dense (0.01, 0.07, 0.12, 0.01 at $N = 30, 100, 300, 1000$) and shaped (0.07, 0.14, 0.25, 0.04). More preferences temporarily made the policy worse. The optimizer is not the cause: 300 and 3000 L-BFGS iterations give identical fits. An exploratory check on the tuning seeds also ruled out the action term $w(s, a)$: removing it does not remove the bump. We leave the mechanism open. *Second*, regret saturates long before the reward is identified. EPIC keeps falling up to $N = 1000$, where it is still 0.31–0.58 from preferences, while regret has been near zero for

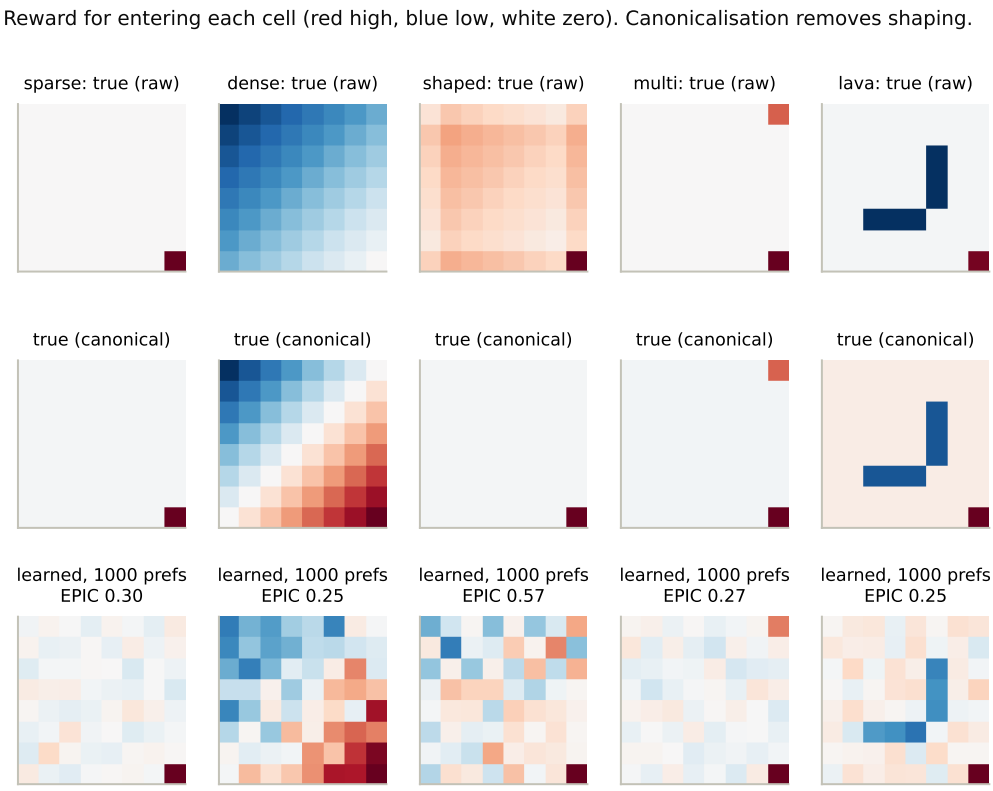


Figure 1: Reward for entering each cell. Top: the raw ground truth. Middle: its EPIC-canonical form (the shaped reward becomes the sparse one). Bottom: canonical rewards learned from 1000 preference pairs (seed 0).

many N. For shaped demonstrations, EPIC barely moves (0.67 at $N = 1000$) even though regret reaches zero by $N = 100$.

3.2 Which distance predicts regret?

Across all main runs, EPIC's Spearman correlation with regret is 0.61 against 0.54 for raw Pearson (difference 0.06, 95% CI [0.02, 0.10]); see Fig. 3. **H2 is supported.** The gap is concentrated where theory says it should be. On the shaped reward, raw Pearson is almost uninformative ($\rho = 0.05$) while EPIC is not ($\rho = 0.46$; difference 0.41 [0.15, 0.67]). Raw correlation is dominated by the shaping term, which does not affect the optimal policy. For the other shapes the two metrics are within a few hundredths of each other. Correlations are weakest for dense ($\rho \approx 0.28$), because regret there is small and noisy at every N.

3.3 When the labeler or the optimizer misbehaves

Myopic labeler. When the labeler discounts by 0.8 but the model assumes 0.95 (Fig. 4, left), regret at $N = 1000$ is 0.192 (sparse), 0.285 (dense), 0.777 (shaped), 0.340 (multi) and 0.899 (lava). With the correct labeler it is at most 0.041. **H3 is not supported as stated**. The worst shapes are lava and shaped, not sparse and lava. And regret is not flat in N: for every shape it is lowest around $N = 10$ (sparse 0.05, lava 0.21) and then rises (lava 0.90 at $N = 1000$). One reading, which we did not test: with little data the ℓ_2 prior dominates the bias, and with more data the model fits the wrong discount confidently. Either way, in this setting more biased feedback was worse than less.

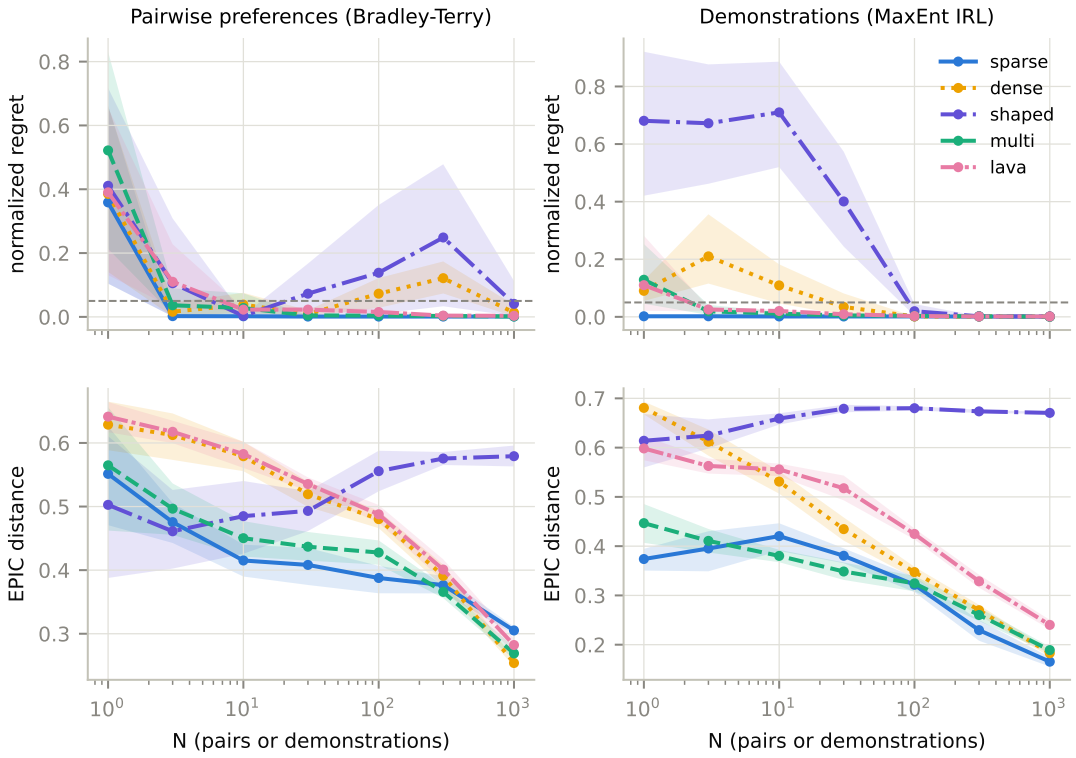


Figure 2: Normalized regret (top) and EPIC distance (bottom) against the amount of feedback. Mean over 10 seeds with 95% bootstrap bands.

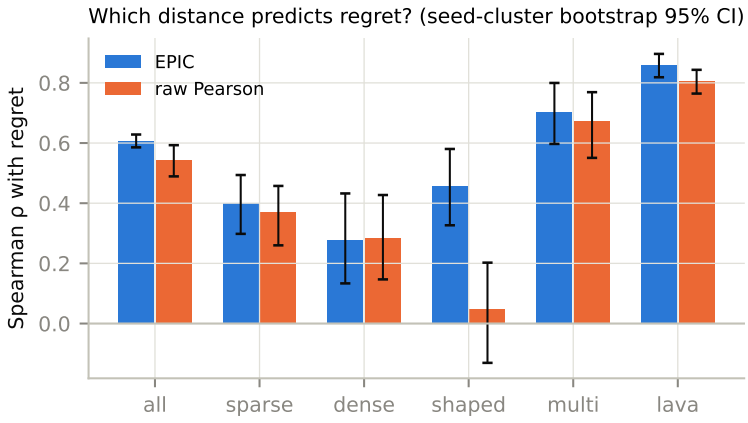


Figure 3: Spearman correlation between each reward distance and normalized regret (seed-cluster bootstrap 95% CI).

Labeler rationality. Lower β hurts the shaped reward far more than the others. At $N = 300$, shaped regret is 0.59 at $\beta = 0.3$, against 0.14 for dense and 0.00 for sparse (Fig. 4, right).

Overoptimization. Following Gao et al. [3], we optimize a learned proxy with a soft-optimal policy at increasing inverse temperature λ and score it on the true reward (Fig. 5). The true return rises and then plateaus, and in most cells it never turns down. The exceptions are the $N = 300$ proxies for dense and shaped, which peak and then lose 0.019 and 0.049 of normalized return. Those proxies also plateau below the ones learned from fewer pairs, which matches the mid-N regret bump in §3.1. In this tabular setting, overoptimization is mild.

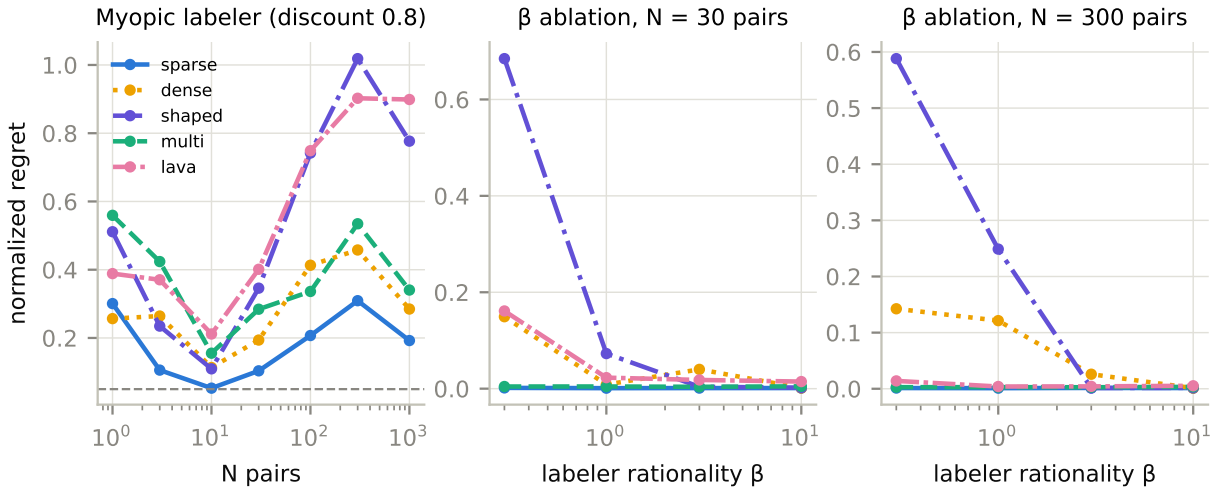


Figure 4: Left: a myopic labeler (discount 0.8). Middle and right: labeler rationality β at $N = 30$ and $N = 300$ pairs.

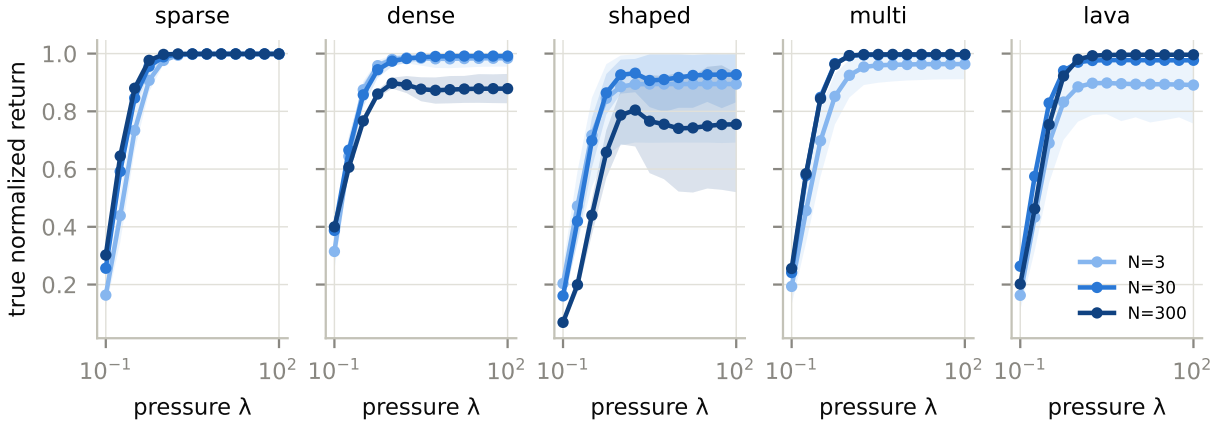


Figure 5: True normalized return of the soft-optimal policy for a learned proxy, against optimization pressure λ.

3.4 Does shaping speed up learning?

As a check on the *use* of shaped rewards, tabular Q-learning (learning rate 0.1, $\epsilon = 0.1$, 50-step episodes) was trained on the sparse, dense or shaped reward and always scored on the sparse task (Fig. 6). The shaped

reward cuts regret fastest in the first few hundred episodes. Dense is slowest early. On the pre-registered criterion (episodes until regret ≤ 0.05), 10/10 dense, 8/10 shaped and only 4/10 sparse seeds crossed the threshold within 3000 episodes (median 955, 935 and 280 episodes among those that did). Sparse training plateaus at a regret of 0.052, just above the line; we did not diagnose why.

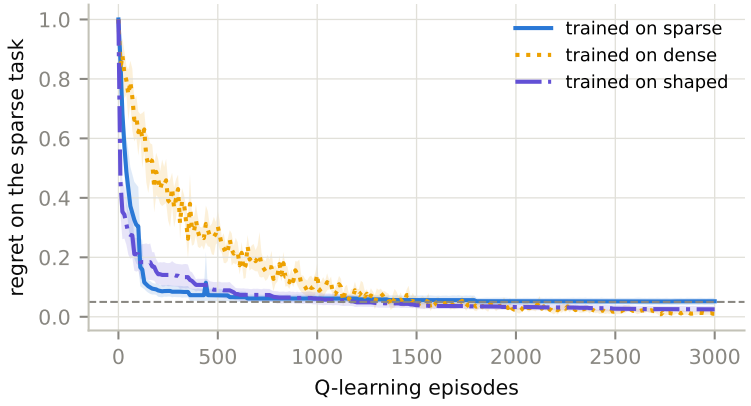


Figure 6: Regret on the sparse task during Q-learning on each training reward (10 seeds, 95% bands).

4 Related work

Preference-based reward learning goes back to paired-comparison models [1] and was popularized for deep RL by Christiano et al. [2]. Knox et al. [5] question the return-based (Boltzmann) labeler that we also assume. Maximum-entropy IRL [10] is our demonstration learner. Potential shaping preserves optimal policies [6], and feedback identifies rewards only up to such transformations [8]. EPIC [4] and its dynamics-aware successor DARD [9] compare rewards modulo shaping. Skalse and Abate [7] analyse how IRL degrades under a misspecified behaviour model, which is the theory behind our myopic-labeler ablation. Gao et al. [3] measure proxy overoptimization in language models; Fig. 5 is a tabular analogue.

5 Limitations

This is one small gridworld with simulated labelers, and the results should not be read as claims about reward learning in general. The compared trajectories come from a generous ϵ-greedy mixture around the optimal policy; uninformed sampling would make every shape harder. The reward model is well specified (except under the myopic ablation), so we test identifiability, not misspecification of the model class. EPIC over the uniform product space counts state–action pairs that no feedback constrains, so its floor stays above zero. A dynamics-aware distance [9] might track regret better. Regret saturates quickly in gridworlds, so N^* falls on the first few grid points for most shapes; a 16×16 check is deferred. The non-monotone regret curves are reported, not explained.

Invented parameters. The grid size and layout, slip probability, γ, H, goal, lava and multi-goal values, step cost, the unit-std rescaling, β, the ϵ-mixture, the demonstrator's temperature, the myopic discount 0.8, the ℓ_2 weight and the 0.05 threshold were all chosen by us, not measured.

Reproducibility. `uv run python -m rewardshapes.run_all` regenerates every run, figure and number in this paper (the environment is pinned in `uv.lock`). All raw runs are included.

References

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